

Mikolaj Kubowicz
10/1/26

Q4.1:

After reading the syntax the debug messages show R4 forming an OSPF adjacency with R3 on GigabitEthernet0/1. The routers first reached the 2-Way state, meaning they successfully exchanged OSPF Hello packets and recognized each other as neighbors. OSPF then performed a DR/BDR election, where R4 was elected DR and R3 was elected BDR on that segment.

The routers then entered the database exchange process. In the EXSTART state, they negotiated which router would act as master and slave for the database exchange. During the EXCHANGE state, they exchanged Database Description (DBD) packets that summarized their OSPF link-state databases. R4 then sent a Link State Request (LS REQ) for information it needed, and R3 responded with Link State Updates (LS UPD). After their databases were synchronized, the adjacency changed from LOADING to FULL, meaning the OSPF neighbor relationship was completely established.

Using wildcard masks instead of only classful network addresses gives the administrator more control over which interfaces participate in OSPF. A wildcard mask can match a specific subnet, such as 10.10.10.0 0.0.0.3, instead of enabling OSPF on an entire classful network. This makes the OSPF configuration more precise and prevents interfaces that should not participate in OSPF from being included.

Q4.2:

R4#show ip protocols

***** IP Routing is NSF aware *****

Routing Protocol is "application"

Sending updates every 0 seconds

Invalid after 0 seconds, hold down 0, flushed after 0

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Maximum path: 32

Routing for Networks:

Routing Information Sources:

Gateway	Distance	Last Update
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Distance: (default is 4)

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 10.10.10.1

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

10.1.1.0 0.0.0.255 area 0

10.10.10.0 0.0.0.3 area 0

Routing Information Sources:

Gateway	Distance	Last Update
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10.20.20.1	110	00:22:00
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192.168.30.1	110	00:17:27
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172.16.41.1	110	00:27:08
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Distance: (default is 110)

R4#

The OSPF router ID is 10.10.10.1

R4 is routing for the following networks in OSPF Area 0:

10.1.1.0 0.0.0.255 area 0

10.10.10.0 0.0.0.3 area 0

Q4.3:

R1#show ip ospf

Routing Process "ospf 1" with ID 172.16.41.1

Start time: 01:40:10.773, Time elapsed: 03:15:29.017

Supports only single TOS(TOS0) routes

Supports opaque LSA

Supports Link-local Signaling (LLS)

Supports area transit capability

Supports NSSA (compatible with RFC 3101)

Supports Database Exchange Summary List Optimization (RFC 5243)

Event-log enabled, Maximum number of events: 1000, Mode: cyclic

Router is not originating router-LSAs with maximum metric

Initial SPF schedule delay 5000 msec

Minimum hold time between two consecutive SPF's 10000 msec

Maximum wait time between two consecutive SPF's 10000 msec

Incremental-SPF disabled

Minimum LSA interval 5 secs

Minimum LSA arrival 1000 msec

LSA group pacing timer 240 secs

Interface flood pacing timer 33 msec

Retransmission pacing timer 66 msec

EXCHANGE/LOADING adjacency limit: initial 300, process maximum 300

Number of external LSA 0. Checksum Sum 0x000000

Number of opaque AS LSA 0. Checksum Sum 0x000000

Number of DCbitless external and opaque AS LSA 0

Number of DoNotAge external and opaque AS LSA 0

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Number of areas transit capable is 0

External flood list length 0

IETF NSF helper support enabled

Cisco NSF helper support enabled

Reference bandwidth unit is 100 mbps

Area BACKBONE(0)

Number of interfaces in this area is 3 (2 loopback)

Area has no authentication

SPF algorithm last executed 03:05:10.126 ago

SPF algorithm executed 6 times

Area ranges are

Number of LSA 7. Checksum Sum 0x024F62

Number of opaque link LSA 0. Checksum Sum 0x000000

Number of DCbitless LSA 0

Number of indication LSA 0

Number of DoNotAge LSA 0

Flood list length 0

OSPF routing process number is 1 and the router ID is 172.16.41.1. The SPF algorithm was last executed about 3 hours, 5 minutes, and 10 seconds ago, and it has run 6 times total. R1 has one OSPF area, Area 0 (the backbone area), and that area currently contains 7 LSAs.

Q.4.3b

R4#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.30.1	1	FULL/BDR	00:00:35	10.10.10.2	GigabitEthernet0/1
10.20.20.1	1	FULL/DROTHER	00:00:35	10.1.1.2	GigabitEthernet0/0
172.16.41.1	1	FULL/BDR	00:00:32	10.1.1.1	GigabitEthernet0/0

R4#

R4 has three OSPF neighbors: R3, R2, and R1. All three are in the FULL state, which means the OSPF neighbor relationships are fully established and the routers have synchronized their routing information. R3 and R1 are shown as BDR, while R2 is shown as DROTHER. The DR and BDR roles are elected on Ethernet networks to help organize OSPF communication, and DROTHER just means that router is not the DR or BDR on that segment.

Q.4.3c.

***Oct 2 00:42:50.660: OSPF-1 ADJ Gi0/1: Interface going Down**

***Oct 2 00:42:50.660: OSPF-1 ADJ Gi0/1: 192.168.30.1 address 10.10.10.2 is dead, state DOWN**

***Oct 2 00:42:50.660: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down: Interface down or detached**

***Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: Nbr 192.168.30.1: Clean-up dbase exchange**

***Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: Neighbor change event**

***Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: DR/BDR election**

***Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: Elect BDR 0.0.0.0**

***Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: Elect DR 10.10.10.1**

***Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: DR: 10.10.10.1 (Id)**

***Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: BDR: none**

*Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: 10.10.10.1 address 10.10.10.1 is dead, state DOWN

*Oct 2 00:42:50.661: OSPF-1 ADJ Gi0/1: Neighbor change event

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: DR/BDR election

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: Elect BDR 0.0.0.0

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: Elect DR 0.0.0.0

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: Elect BDR 0.0.0.0

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: Elect DR 0.0.0.0

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: DR: none

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: BDR: none

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: Flush network LSA immediately

*Oct 2 00:42:50.662: OSPF-1 ADJ Gi0/1: Remember old DR 10.10.10.1 (id)

*Oct 2 00:42:50.663: OSPF-1 ADJ Gi0/1: Interface going Up

*Oct 2 00:42:50.666: OSPF-1 ADJ Gi0/1: 2 Way Communication to 192.168.30.1, state 2WAY

*Oct 2 00:42:50.666: OSPF-1 ADJ Gi0/1: Nbr 192.168.30.1: Prepare dbase exchange

*Oct 2 00:42:50.666: OSPF-1 ADJ Gi0/1: Send DBD to 192.168.30.1 seq 0x1109 opt 0x52 flag 0x7 len 32

*Oct 2 00:42:50.669: OSPF-1 ADJ Gi0/1: Rcv DBD from 192.168.30.1 seq 0x10DB opt 0x52 flag 0x7 len 32 mtu 1500 state EXSTART

*Oct 2 00:42:50.669: OSPF-1 ADJ Gi0/1: NBR Negotiation Done. We are the SLAVE

*Oct 2 00:42:50.669: OSPF-1 ADJ Gi0/1: Nbr 192.168.30.1: Summary list built, size 7

*Oct 2 00:42:50.669: OSPF-1 ADJ Gi0/1: Send DBD to 192.168.30.1 seq 0x10DB opt 0x52 flag 0x2 len 172

*Oct 2 00:42:50.672: OSPF-1 ADJ Gi0/1: Rcv DBD from 192.168.30.1 seq 0x10DC opt 0x52 flag 0x1 len 52 mtu 1500 state EXCHANGE

*Oct 2 00:42:50.673: OSPF-1 ADJ Gi0/1: Exchange Done with 192.168.30.1

*Oct 2 00:42:50.673: OSPF-1 ADJ Gi0/1: Synchronized with 192.168.30.1, state FULL

*Oct 2 00:42:50.673: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

*Oct 2 00:42:50.673: OSPF-1 ADJ Gi0/1: Send DBD to 192.168.30.1 seq 0x10DC opt 0x52 flag 0x0 len 32

R4#

When I changed the R3–R4 link to point-to-point, the existing OSPF adjacency first went from FULL to DOWN because the interface's OSPF network type changed. R4 then removed the old DR/BDR information and flushed the old network LSA. After that, the interface came back up and R4 and R3 formed a new adjacency. They reached the 2-Way state, exchanged database information through the EXSTART and EXCHANGE process, and then returned to the FULL state. Because the link is now point-to-point, OSPF no longer needs a DR or BDR election for that link.

Q.4.3d.

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.30.1	0	FULL/ -	00:00:33	10.10.10.2	GigabitEthernet0/1
10.20.20.1	1	FULL/DROTHER	00:00:36	10.1.1.2	GigabitEthernet0/0
172.16.41.1	1	FULL/BDR	00:00:38	10.1.1.1	GigabitEthernet0/0

R4#

R4 still has three OSPF neighbors, and all three adjacencies are in the FULL state. R3, with router ID 192.168.30.1, now shows FULL/- instead of FULL/BDR because the R3–R4 link was changed to an OSPF point-to-point network. Point-to-point links only connect two routers, so OSPF does not need to elect a DR or BDR on that link. R1 and R2 are still on the shared Ethernet network, so they continue to use the normal DR/BDR/DROTHER roles.

Q.4.4a.

R1#show ip route 192.168.30.1

Routing entry for 192.168.30.1/32

Known via "ospf 1", distance 110, metric 202, type intra area

Last update from 10.1.1.4 on GigabitEthernet0/0, 00:24:01 ago

Routing Descriptor Blocks:

10.1.1.4, from 192.168.30.1, 00:24:01 ago, via GigabitEthernet0/0

Route metric is 202, traffic share count is 1

*** 10.1.1.2, from 192.168.30.1, 03:59:17 ago, via GigabitEthernet0/0**

Route metric is 202, traffic share count is 1

R1#

R2#show ip route 192.168.30.1

Routing entry for 192.168.30.1/32

Known via "ospf 1", distance 110, metric 201, type intra area

Last update from 10.20.20.2 on GigabitEthernet0/1, 03:59:09 ago

Routing Descriptor Blocks:

*** 10.20.20.2, from 192.168.30.1, 03:59:09 ago, via GigabitEthernet0/1**

Route metric is 201, traffic share count is 1

R2#

From R2, the OSPF cost to R3's Loopback0 network is 201, and the next hop is 10.20.20.2, which is R3. From R1, the cost is 202. R1 has two equal-cost next hops, 10.1.1.2 through R2 and 10.1.1.4 through R4, so OSPF can use either path to reach R3's loopback network.

Q4.4b

R2#show ip route 192.168.30.1

Routing entry for 192.168.30.1/32

Known via "ospf 1", distance 110, metric 202, type intra area

Last update from 10.1.1.4 on GigabitEthernet0/0, 00:00:20 ago

Routing Descriptor Blocks:

*** 10.1.1.4, from 192.168.30.1, 00:00:20 ago, via GigabitEthernet0/0**

Route metric is 202, traffic share count is 1

R2#

R1#show ip route 192.168.30.1

Routing entry for 192.168.30.1/32

Known via "ospf 1", distance 110, metric 202, type intra area

Last update from 10.1.1.4 on GigabitEthernet0/0, 00:30:05 ago

Routing Descriptor Blocks:

*** 10.1.1.4, from 192.168.30.1, 00:30:05 ago, via GigabitEthernet0/0**

Route metric is 202, traffic share count is 1

R1#

After changing the OSPF cost on the R2–R3 link to 10000, R2 now reaches R3's Loopback0 with a cost of 202 and a next hop of 10.1.1.4, which is R4. Before the change, R2 used the direct path to R3 with a cost of 201. R1 still has a cost of 202, but now its only next hop is 10.1.1.4 through R4. Before the change, R1 had two equal-cost paths through R2 and R4. Increasing the cost of the R2–R3 link caused OSPF to stop preferring that path and use R4 instead.

The key idea is that setting the interface cost to 10000 did not make the final route cost 10000, because OSPF found a cheaper alternate route through R4.

Q4.4c

R1#show ip route 192.168.30.1

Routing entry for 192.168.30.1/32

Known via "ospf 1", distance 110, metric 10002, type intra area

Last update from 10.1.1.2 on GigabitEthernet0/0, 00:00:28 ago

Routing Descriptor Blocks:

*** 10.1.1.2, from 192.168.30.1, 00:00:28 ago, via GigabitEthernet0/0**

Route metric is 10002, traffic share count is 1

R1#

R2#show ip route 192.168.30.1

Routing entry for 192.168.30.1/32

Known via "ospf 1", distance 110, metric 10001, type intra area

Last update from 10.20.20.2 on GigabitEthernet0/1, 00:00:16 ago

Routing Descriptor Blocks:

*** 10.20.20.2, from 192.168.30.1, 00:00:16 ago, via GigabitEthernet0/1**

Route metric is 10001, traffic share count is 1

R2#

I changed the OSPF cost on the R3-R4 link to 20000 on R4 G0/1 and R3 G0/0. Since the R2-R3 link still has a cost of 10000, OSPF now prefers the lower-cost path through R2. From R2, the cost to R3's Loopback0 is now 10001 with next hop 10.20.20.2. From R1, the cost is now 10002 with next hop 10.1.1.2, which is R2. This shows that OSPF chooses the path with the lower total cost.

R4(config)#interface g0/1

R4(config-if)#ip ospf cost 20000

R3(config)#interface g0/0

R3(config-if)#ip ospf cost 20000

EXTRA CREDIT

Starting at R1, the initial cost is 0. R1 can reach both R2 and R4 across the shared network with a cost of 1, so both are the first lowest-cost choices. From R2, the path to R3 adds a cost of 10000, giving a total cost of 10001 to R3. From R4, the path to R3 adds a cost of 20000, giving a total cost of 20001. Therefore, Dijkstra selects the path through R2 because 10001 is lower than 20001. The shortest path from r1 to r3 is R1 to R2 to R3

The cost to reach R3 is 10001. R3's Loopback0 adds another cost of 1, so the total OSPF cost from R1 to 192.168.30.1 is 10002

HOURS TO COMPLETE 3